

JIMMY LAKE, SK, Canada

WMO: 710230

Lat: 54.908N

Lon: 109.962W

Elev: 2090

StdP: 13.62

Time Zone: -6.00 (W06)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-20.6	-33.9	0.9	-24.9	-29.3	1.1	-19.9	23.2	32.6	20.3	27.6	4.8	320	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	79.9	63.0	76.9	61.7	74.2	60.1	65.3	75.5	63.4	73.2	61.8	71.0	9.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	88.5	69.7	59.6	82.5	67.5	58.0	77.8	65.7	31.4	75.3	29.9	73.3	28.7	71.3	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.8	16.4	14.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-31.8	85.4	5.8	1.9	-36.0	86.8	-39.4	87.9	-42.7	89.0	-46.9	90.4
			-31.3	69.2	5.0	1.7	-34.9	70.4	-37.8	71.4	-40.6	72.3	-44.2	73.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	6.3	8.1	21.1	35.8	49.2	57.6	62.5	59.5	50.5	37.2	20.8	7.2
	DBStd	23.18	16.24	13.98	14.45	10.21	8.82	6.00	4.87	6.05	7.95	9.05	11.97	14.06
	HDD50	6693	1355	1174	896	433	123	10	0	5	89	406	875	1327
	HDD65	11081	1820	1594	1360	875	493	232	108	185	436	861	1325	1792
	CDD50	1145	0	0	0	8	98	237	387	301	104	10	0	0
	CDD65	58	0	0	0	0	2	9	29	16	2	0	0	0
	CDH74	622	0	0	0	2	45	107	263	162	42	1	0	0
	CDH80	70	0	0	0	0	6	12	34	14	4	0	0	0
	WSAvg	6.8	6.4	6.5	7.4	7.8	7.7	6.7	6.5	6.2	6.6	7.0	6.7	5.8
	PrecAvg	16.9	0.8	0.6	0.7	1.2	1.4	2.7	3.1	2.3	1.7	0.9	0.7	0.9
(15) Precipitation	PrecMax	21.1	1.5	1.1	1.3	3.0	2.5	6.0	4.6	5.6	3.6	1.8	1.6	2.0
	PrecMin	12.1	0.3	0.2	0.0	0.2	0.2	1.3	1.3	0.8	0.5	0.3	0.2	0.3
	PrecStd	2.3	0.4	0.3	0.4	0.6	0.7	1.1	1.0	1.2	0.9	0.4	0.4	0.4
	PrecStd	2.3	0.4	0.3	0.4	0.6	0.7	1.1	1.0	1.2	0.9	0.4	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	43.8	53.1	70.2	79.8	81.7	83.4	81.8	79.3	68.8	49.6	39.7
		MCWB	36.0	35.4	41.0	50.5	57.4	61.0	66.0	65.2	62.3	53.8	40.6	33.9
	2%	DB	38.5	37.6	47.3	62.3	73.2	76.9	80.0	78.4	73.4	59.9	43.9	35.0
		MCWB	33.5	31.3	37.5	46.0	54.2	59.7	64.6	63.4	58.9	48.8	37.7	31.6
	5%	DB	34.2	33.0	43.2	56.9	69.3	73.6	76.7	75.2	68.3	54.4	39.5	30.6
		MCWB	31.1	28.7	34.9	42.6	52.2	58.0	62.8	61.9	56.8	45.1	34.6	28.1
	10%	DB	29.4	28.7	39.7	52.1	65.5	70.0	74.0	72.1	63.8	50.3	35.4	26.5
		MCWB	26.9	25.6	33.0	40.4	50.0	56.0	61.4	60.7	53.9	42.7	31.9	24.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	36.1	41.5	50.9	58.7	63.5	68.9	67.6	63.7	54.4	41.9	35.1
		MCDB	43.3	43.5	51.3	68.5	75.9	75.6	78.6	78.4	76.0	66.9	47.7	38.1
	2%	WB	34.1	32.1	38.1	46.6	56.0	61.3	66.1	64.9	60.5	49.6	38.6	31.7
		MCDB	38.3	36.4	46.6	61.2	69.8	72.6	76.0	75.1	70.7	58.7	43.5	35.5
	5%	WB	31.4	29.1	35.5	43.3	53.7	59.8	64.0	63.1	57.7	46.2	35.6	28.3
		MCDB	34.8	33.4	42.7	55.8	66.3	69.8	73.4	73.0	67.1	53.2	39.6	30.9
	10%	WB	27.6	26.0	33.1	40.7	51.6	58.1	62.3	61.6	55.0	43.3	32.4	24.6
		MCDB	30.3	29.2	39.0	51.3	62.9	66.6	70.9	70.4	62.2	49.7	35.3	26.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	18.2	18.3	18.7	20.8	18.6	18.7	18.5	17.9	14.4	12.9	14.4
		MCDBR	18.8	21.2	21.3	27.0	26.4	24.1	23.3	23.6	25.4	22.3	16.5	19.2
		MCWBR	15.1	16.0	14.1	15.8	13.3	11.0	11.0	11.2	13.2	13.0	11.5	15.9
	5% WB	MCDBR	18.5	21.5	20.6	26.5	23.8	20.1	20.8	21.8	24.5	21.5	16.5	18.9
		MCWBR	15.0	16.9	13.9	15.6	12.5	10.3	10.7	11.0	13.1	13.1	12.1	15.9
(40) Clear-Sky Solar Irradiance	taub		0.239	0.268	0.273	0.314	0.333	0.352	0.355	0.350	0.306	0.276	0.244	0.214
	taud		2.333	2.307	2.434	2.462	2.452	2.410	2.448	2.421	2.551	2.601	2.466	2.352
	Ebn at Noon		234	263	292	291	289	284	280	275	277	264	235	223
	Edh at Noon		16	24	27	30	33	35	33	32	24	18	14	12
(44) All-Sky Solar Radiation	RadAvg		258	583	1072	1476	1777	1791	1804	1483	1034	572	257	177
	RadStd		25	43	71	116	168	134	111	121	114	94	35	16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page